ONEDRAFT

CAD-AI — Aria Powered

Developer Platform, SDK & Extension Architecture

Version 1.0

Document: OneDraft Developer Platform, SDK & Extension Architecture Specification
System: OneDraft CAD-AI
Intelligence Layer: Aria
Document Class: Architectural & Implementation Stack Specification
Status: Architectural Specification
Predecessor: Disaster Testing, Security Testing & Operational Readiness Architecture Specification
Successor: Product Configuration & Administrative Control Plane Architecture Specification

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1. PURPOSE

The Developer Platform, SDK & Extension Architecture establishes the supported development interfaces through which developers, organizations and approved external systems may extend, integrate with and automate OneDraft.

The purpose of this subsystem is to provide stable, governed and versioned mechanisms for interacting with OneDraft without requiring direct access to internal implementation details.

The architecture shall support Python and TypeScript clients, SDKs, plugins, external integrations, automation, webhooks and future third-party extensions while preserving security, tenant isolation, API contracts and platform integrity.

2. ARCHITECTURAL ROLE

The Developer Platform forms the controlled extensibility boundary of OneDraft.

It connects public APIs, SDKs, developer tooling, plugins, event interfaces and external integrations to the internal OneDraft platform.

Primary responsibility: Provide supported mechanisms for programmatic interaction and controlled extension.

Primary inputs: API requests, SDK operations, plugin requests, events, webhooks and integration commands.

Primary outputs: API responses, SDK objects, events, artifacts, job references and extension results.

Authority: The Developer Platform shall operate within the API, authorization, tenant, lifecycle and domain contracts established by preceding architectures.

3. EXTENSION MODEL

The OneDraft extension architecture shall follow:

```
DEVELOPER
   ↓
SDK / API / PLUGIN
   ↓
API GATEWAY
   ↓
AUTHENTICATION
   ↓
AUTHORIZATION
   ↓
TENANT CONTEXT
   ↓
APPLICATION SERVICES
   ↓
DOMAIN / COMPUTATIONAL SERVICES
```

Extensions shall interact with supported contracts rather than bypassing the application architecture.

4. DEVELOPER INTERFACES

Supported developer interfaces shall include:

REST API

Python SDK

TypeScript SDK

Webhooks

Event Interfaces

Plugin Interfaces

Command Interfaces

Artifact Interfaces

Future Extension APIs

Each interface shall have defined compatibility and security requirements.

5. API FOUNDATION

The OneDraft API shall remain the foundational developer interface.

The API shall expose controlled access to:

Organizations

Projects

Requirements

Models

Geometry

Validation

Compliance

Drawings

Reviews

Redlines

Jobs

Artifacts

Aria

Billing

Integrations

Only capabilities explicitly included in the public contract shall be externally accessible.

6. API VERSIONING

Developer-facing APIs shall be versioned.

Representative structure:

```
/api/v1/...
```

Breaking changes shall require controlled version evolution.

Deprecated endpoints shall have documented retirement policies.

7. SDK ARCHITECTURE

SDKs shall provide typed and ergonomic access to the public OneDraft API.

The SDK architecture shall separate:

Transport

Authentication

Request Construction

Serialization

Response Models

Error Handling

Retry Handling

Pagination

Async Operations

This separation shall allow SDK implementations to evolve without changing domain semantics.

8. PYTHON SDK

The Python SDK shall support Python-based integrations and automation.

Representative capabilities include:

Project Management

Model Operations

Validation

Compliance Queries

Drawing Generation

Job Monitoring

Artifact Management

Aria Commands

Webhook Handling

The Python SDK shall use typed domain representations where practical.

9. TYPESCRIPT SDK

The TypeScript SDK shall support browser and server-side JavaScript/TypeScript applications.

It shall provide:

Typed API Clients

Domain Models

Request Builders

Error Types

Job Clients

Artifact Clients

Event Utilities

The SDK shall remain compatible with the supported OneDraft frontend architecture.

10. SDK DOMAIN MODELS

SDK models shall correspond to stable public domain contracts.

Representative models include:

Organization

User

Project

Requirement

Site

Building

Level

Space

Element

ModelVersion

Revision

Drawing

Sheet

Annotation

Redline

Job

Artifact

AriaCommand

SDK models shall not expose unstable internal implementation structures unnecessarily.

11. AUTHENTICATION

Developer clients shall authenticate using supported mechanisms.

These may include:

OAuth/OIDC

API Keys

Service Credentials

Signed Tokens

Authentication credentials shall be scoped and revocable.

SDKs shall not embed platform secrets into distributed client applications.

12. API KEY MANAGEMENT

Organization API keys shall be:

Unique

Scoped

Revocable

Rotatable

Auditable

API keys shall be stored securely and shall not be retrievable in plaintext after creation where platform architecture permits.

13. SERVICE CREDENTIALS

Server-to-server integrations shall use dedicated service credentials.

Service credentials shall be separate from individual user credentials.

Credential lifecycle shall support:

Creation

Rotation

Revocation

Expiration

Audit

14. TENANT ISOLATION

Developer access shall inherit the Multi-Tenant Organization & Project Isolation architecture.

Every request shall operate within an authorized organization context.

External developers shall not obtain cross-organization access by manipulating project or resource identifiers.

15. PROJECT SCOPING

API credentials and integrations may be scoped to specific projects.

A project-scoped credential shall not automatically gain organization-wide access.

Project scope shall be enforced server-side.

16. PERMISSION SCOPING

Developer credentials shall support least-privilege permissions.

Representative permissions include:

projects:read

projects:write

models:read

models:write

drawings:read

drawings:generate

artifacts:read

artifacts:write

jobs:read

aria:execute

Permissions shall correspond to actual OneDraft authorization capabilities.

17. SDK ERROR MODEL

SDKs shall provide structured error types.

Errors shall expose appropriate information such as:

Error Code

HTTP Status

Request Identifier

Correlation Identifier

Validation Details

SDKs shall avoid exposing internal stack traces or sensitive infrastructure information.

18. RETRY BEHAVIOR

SDKs may provide controlled retry behavior.

Retries shall consider:

HTTP Status

Idempotency

Retry-After

Request Type

Network Failure

SDKs shall not automatically retry unsafe mutations without appropriate idempotency protection.

19. PAGINATION

Collection APIs shall support deterministic pagination.

Pagination may use:

Cursor-Based Pagination

Offset-Based Pagination

Cursor-based pagination shall be preferred for large and frequently changing datasets where appropriate.

20. FILTERING & SORTING

Public APIs may support controlled filtering and sorting.

Filters shall be:

Explicitly Defined

Validated

Tenant-Aware

Resource-Aware

Clients shall not submit arbitrary database query expressions.

21. ASYNCHRONOUS SDK OPERATIONS

SDKs shall provide first-class support for long-running operations.

Representative workflow:

```
CLIENT
  ↓
CREATE JOB
  ↓
JOB ID
  ↓
POLL / EVENT
  ↓
COMPLETION
  ↓
RESULT
```

SDKs may provide both polling and event-driven completion mechanisms.

22. JOB CLIENT

The SDK shall provide a common job interface.

Representative operations:

Create

Get

Cancel

Retry, where permitted.

Wait

Get Result

Job state shall correspond to the canonical OneDraft job lifecycle.

23. WEBHOOKS

OneDraft shall support webhooks for relevant asynchronous events.

Webhook events may include:

Project Updated

Model Version Created

Validation Completed

Drawing Generated

Export Completed

Job Completed

Review Submitted

Artifact Created

Webhook delivery shall be tenant-scoped.

24. WEBHOOK SECURITY

Webhook requests shall use authentication or signing mechanisms.

The receiver shall be able to verify:

Event Origin

Event Integrity

Event Identifier

Timestamp

Delivery Attempt

Replay protection shall be supported where appropriate.

25. WEBHOOK DELIVERY

Webhook delivery shall support:

Retry

Backoff

Dead-Letter Handling

Delivery Tracking

Failure Detection

Webhook consumers shall be responsible for idempotent event handling.

26. EVENT INTERFACE

The event architecture shall expose approved domain events to authorized integrations.

Events shall be versioned.

Representative structure:

```
Event
 ├── Event ID
 ├── Event Type
 ├── Event Version
 ├── Organization ID
 ├── Project ID
 ├── Resource ID
 ├── Timestamp
 └── Payload
```

Only explicitly public events shall be exposed externally.

27. PLUGIN ARCHITECTURE

OneDraft may support controlled plugins.

Plugins may extend:

User Interface

Drawing Tools

Import Pipelines

Export Pipelines

Validation Rules

Data Integrations

Project Automation

Plugins shall execute through defined extension contracts.

28. PLUGIN ISOLATION

Plugins shall not receive unrestricted platform access.

Plugin execution shall be constrained by:

Permissions

Tenant Scope

API Scope

Resource Scope

Execution Environment

Network Policy

Plugins shall not bypass authorization through internal APIs.

29. PLUGIN LIFECYCLE

The plugin lifecycle shall include:

```
REGISTER
   ↓
VALIDATE
   ↓
APPROVE
   ↓
INSTALL
   ↓
CONFIGURE
   ↓
ENABLE
   ↓
EXECUTE
   ↓
DISABLE
   ↓
REMOVE
```

Plugin lifecycle operations shall be auditable.

30. PLUGIN MANIFEST

Plugins shall provide a manifest describing:

Plugin Identifier

Version

Developer

Capabilities

Required Permissions

Supported API Versions

Dependencies

Configuration Requirements

Entry Points

The manifest shall be validated before installation.

31. EXTENSION PERMISSIONS

Extensions shall declare required permissions before activation.

Permissions shall follow least privilege.

A plugin requesting access to unrelated organizational resources shall not be approved automatically.

32. EXTENSION COMPATIBILITY

Extensions shall declare supported platform and API versions.

Compatibility shall be evaluated during:

Installation

Upgrade

Runtime

Breaking platform changes shall be detectable before an incompatible extension is activated.

33. EXTENSION VERSIONING

Plugins and SDKs shall use semantic or equivalent controlled versioning.

Versions shall identify:

Major Changes

Minor Features

Patch Fixes

Versioning policy shall remain synchronized with API compatibility.

34. SDK RELEASE MANAGEMENT

SDK releases shall be generated and published through the CI/CD architecture.

Each release shall identify:

SDK Version

API Version

Supported Runtime Versions

Dependency Versions

Release Date

Compatibility Status

SDK releases shall pass automated tests before publication.

35. CODE GENERATION

Where practical, SDK models and clients may be generated from the authoritative OpenAPI specification.

Generated components shall remain distinguishable from manually maintained implementation code.

The generated API surface shall be tested against the actual service contract.

36. CLI

OneDraft may provide a developer command-line interface.

Representative commands may include:

```
onedraft projects list
onedraft projects get
onedraft jobs get
onedraft jobs wait
onedraft artifacts download
onedraft validate
onedraft export
onedraft aria execute
```

CLI operations shall use the same API and authorization model as SDK clients.

37. LOCAL DEVELOPMENT

The Developer Platform shall provide local development support where practical.

Development tooling may provide:

Local API

Test Database

Mock Services

Example Projects

Webhook Receiver

SDK Fixtures

CLI

Local tooling shall not contain production credentials.

38. SANDBOX ENVIRONMENT

OneDraft may provide a developer sandbox.

The sandbox shall provide isolated:

Organizations

Projects

API Credentials

Artifacts

Jobs

Events

Sandbox data shall remain separate from production data.

39. TESTING EXTENSIONS

Developer extensions shall be tested against controlled environments.

Testing shall include:

API Compatibility

Permission Enforcement

Tenant Isolation

Webhook Handling

Failure Handling

Version Compatibility

Performance

Security

40. EXTERNAL INTEGRATIONS

External integrations may connect through:

REST APIs

SDKs

Webhooks

Event Interfaces

File Exchange

OAuth/OIDC

Integrations shall remain explicitly authorized.

41. INTEGRATION REGISTRY

OneDraft shall maintain metadata for registered integrations.

Records may include:

Integration ID

Organization

Developer

Version

Permissions

Credentials

Status

Configuration

Last Activity

Integration lifecycle changes shall be auditable.

42. EXTENSION OBSERVABILITY

Extension execution shall generate appropriate telemetry.

Metrics may include:

Invocation Count

Latency

Error Rate

Rate-Limit Events

Webhook Failures

Job Failures

Resource Consumption

Observability shall preserve tenant boundaries.

43. EXTENSION RESOURCE LIMITS

Extensions shall operate within controlled resource limits.

Limits may apply to:

CPU

Memory

Execution Time

API Calls

Storage

Network Requests

Concurrent Jobs

Resource limits shall protect platform availability.

44. EXTENSION SECURITY

Extension security shall include:

Permission Review

Credential Isolation

Tenant Isolation

Input Validation

Dependency Review

Code Integrity

Runtime Isolation

Audit Logging

Extensions shall be treated as untrusted or semi-trusted components unless explicitly established otherwise.

45. DEVELOPER DOCUMENTATION

The platform shall provide documentation covering:

API Reference

SDK Reference

Authentication

Authorization

Webhooks

Events

Plugins

Examples

Error Handling

Versioning

Migration Guides

Documentation shall correspond to supported API versions.

46. DEVELOPER SUPPORT

Developer-facing support mechanisms may include:

API Status Information

Documentation

SDK Issue Reporting

Integration Diagnostics

Webhook Delivery Logs

Migration Guidance

Support access shall preserve organization and project isolation.

47. ARCHITECTURAL INVARIANTS

The following invariants shall govern the Developer Platform, SDK & Extension Architecture.

API Authority: Public integrations shall use supported API contracts.

Tenant Isolation: Developer access shall never bypass organization or project boundaries.

Least Privilege: Extensions shall receive only required permissions.

Version Control: Public interfaces shall be versioned.

Compatibility: SDKs and plugins shall declare supported versions.

Credential Security: Developer credentials shall be protected, scoped and revocable.

Idempotency: Retryable operations shall preserve domain correctness.

Auditability: Extension activity shall remain traceable.

Resource Protection: Extensions shall not exhaust shared platform resources.

No Internal Bypass: Public extensions shall not rely upon private internal service interfaces.

Deterministic Domain Control: Extensions shall remain subject to OneDraft domain and validation rules.

48. IMPLEMENTATION REQUIREMENTS

The implementation shall provide:

Public API

Python SDK

TypeScript SDK

Authentication Framework

Authorization Integration

API Versioning

Job Client

Webhook System

Event Interface

Plugin Framework

Extension Manifest

Permission Model

Developer CLI

Sandbox Support

Integration Registry

Developer Documentation

SDK Release Pipeline

Extension Testing Framework

These components shall integrate with the API Gateway, Identity, Authorization, Tenant Isolation, Runtime, Job, Event, Artifact, AI and CI/CD architectures.

49. EXTENSION ROADMAP

The initial implementation shall prioritize stable API and SDK capabilities.

Subsequent extension capabilities may include:

Advanced Plugins

Custom Validation Providers

Custom Drawing Generators

External BIM Connectors

Survey Integrations

Enterprise Automation

Third-Party Model Providers

Developer Marketplace

Future extensions shall remain subordinate to the core OneDraft authorization, validation and security architecture.

50. COMPLETION STATUS

The Developer Platform, SDK & Extension Architecture establishes the supported extensibility model for OneDraft.

The subsystem defines public APIs, Python and TypeScript SDKs, authentication, authorization, tenant scoping, API keys, service credentials, SDK models, error handling, retries, pagination, asynchronous jobs, webhooks, event interfaces, plugins, extension permissions, manifests, compatibility, versioning, CLI tooling, local development, sandbox environments, external integrations, integration registration, extension observability, resource limits, security, developer documentation and extension governance.

The architecture establishes a controlled development ecosystem through which OneDraft can be automated, integrated and extended without compromising the internal domain model, security boundaries or tenant isolation.

The next architectural layer establishes the Product Configuration & Administrative Control Plane, defining how platform administrators manage controlled system settings, feature flags, jurisdiction configuration, templates, product behavior and administrative policies.

Status: ARCHITECTURALLY DEFINED

Next Document: Product Configuration & Administrative Control Plane Architecture Specification

Overall Architecture: OneDraft End-to-End System Integration